What do you want to learn?

ALICE L WALTON SCHOOL OF MEDICINE

ASCVD–PREVENT Risk Score in Dyslipidemia Management

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Authors: Diptiman Bose, Phil Riley, Trager Hintze
Editors: Jordan Finn, Russell Johnson

Learning Objectives (4)

After completing this brick, you will be able to:

1. Define the purpose and components of the ASCVD-PREVENT risk score.
2. Describe the clinical factors used in calculating the ASCVD-PREVENT risk score.
3. Explain how the ASCVD-PREVENT risk score estimates cardiovascular risk in patients with dyslipidemia
4. Apply the ASCVD-PREVENT risk score to calculate individual patient risk for cardiovascular events

CASE CONNECTION

RJ, 52-year-old male presents at primary care clinic with elevated cholesterol detected during routine health check. He is concerned about his long-term cardiovascular risk and asks how his results will impact medical decisions. SM is a third-year medical student, working alongside the physician to understand risk stratification tools for dyslipidemia.

What factors are included in the ASCVD-PREVENT risk score for RJ, and how does this score assist in clinical decision-making for his dyslipidemia? Consider your answers as you read, and we'll revisit RJ at the end of the brick.

GO TO CONCLUSION ↓

What Is the Purpose and What Are the Components of the ASCVD–PREVENT Risk Score? (LO#1)

The ASCVD-PREVENT (Atherosclerotic Cardiovascular Disease-Predicting Risk of Vascular Events and Net Treatment Effects) risk score represents a significant advancement in cardiovascular risk assessment. This risk prediction tool was developed to address limitations in previous risk calculators (the Pooled Cohort Equation; aka the ASCVD Score) by providing more personalized risk estimates and treatment benefit projections for patients with dyslipidemia. Unlike earlier models that focused primarily on initial risk assessment, the ASCVD-PREVENT score integrates both risk prediction and potential treatment benefit into a comprehensive framework for clinical decision-making.

Score Objective

The primary objective of the ASCVD-PREVENT risk score is to provide clinicians with a reliable method to estimate a patient's 10-year risk of experiencing atherosclerotic cardiovascular disease events, including myocardial infarction, stroke, and cardiovascular death. Beyond simply calculating risk, this tool aims to guide evidence-based treatment decisions by quantifying the potential benefit of preventive interventions, particularly lipid-lowering therapies. The score was specifically designed to help clinicians identify patients most likely to benefit from statin therapy and other lipid-lowering interventions.

Additionally, the ASCVD-PREVENT score addresses limitations of previous risk calculators by incorporating more contemporary population data and a broader range of risk factors. It provides more accurate risk stratification across diverse patient populations, including those with varying ethnic backgrounds and comorbidities. This enhanced precision helps clinicians and patients engage in more informed shared decision-making regarding preventive cardiovascular care.

Score Components

The ASCVD-PREVENT risk score incorporates multiple components that collectively provide a comprehensive assessment of cardiovascular risk. These components can be broadly categorized into non-modifiable risk factors, modifiable risk factors, and laboratory parameters. The integration of these diverse elements allows for a nuanced evaluation of individual risk profiles and potential treatment benefits.

The key components clude:

- **Non-modifiable factors**: Age, sex, race/ethnicity, and family history of premature cardiovascular disease
- **Modifiable lifestyle factors**: Smoking status, physical activity level, and body mass index
- **Comorbidities**: Diabetes status, hypertension, and chronic kidney disease
- **Laboratory parameters**: Total cholesterol, HDL cholesterol, LDL cholesterol, triglycerides, and high-sensitivity C-reactive protein (hsCRP)
- **Treatment variables**: Current use of antihypertensive medications, aspirin, and lipid-lowering therapies

ACTIVE LEARNING QUESTION

Which of the following are components of the ASCVD-PREVENT risk score?

A. ☑ Total cholesterol
B. ☑ HDL cholesterol
C. ☑ LDL-C
D. ☑ Triglycerides
E. ☐ HbA1c
F. ☐ Smoking status
G. ☑ hsCR

Explanation

Hemoglobin A1c and smoking status are not included in the score calculation

By incorporating this comprehensive set of variables, the ASCVD-PREVENT score provides a more personalized risk assessment than previous models. The score's algorithm weighs these factors differently based on their relative contribution to cardiovascular risk, with certain variables (such as age, LDL cholesterol, and diabetes status) carrying particularly significant weight in the final calculation.

Which Clinical Factors Are Used in Calculating the ASCVD–PREVENT Risk Score? (LO#2)

The ASCVD-PREVENT risk score calculation relies on a comprehensive set of clinical factors that have demonstrated significant associations with cardiovascular outcomes in large epidemiological studies. These factors are systematically assessed and weighted within the algorithm to generate an individualized risk estimate. Understanding these clinical inputs is essential for healthcare providers to accurately implement the risk score in practice and interpret its results ap

Demographics and History

Demographic factors and patient history elements form the foundation of the ASCVD-PREVENT risk assessment. Age is one of the most powerful predictors of cardiovascular risk, with the score algorithm applying progressively higher risk weightings as age increases, particularly after 40 years. Sex is another critical factor, as males generally demonstrate higher baseline cardiovascular risk compared to females of the same age. The algorithm accounts for this difference by applying sex-specific risk coefficients.

Race and ethnicity significantly influence cardiovascular risk profiles, with the ASCVD-PREVENT score specifically accounting for these differences. The model includes separate risk calculations for White, Black, Hispanic, and Asian populations, addressing the known variations in cardiovascular disease patterns across these groups. For instance, Black individuals often have higher stroke risk at equivalent risk factor levels compared to White individuals, while Asian populations may have different patterns of lipid-related risk.

What demographic factors are crucial in ASCVD-PREVENT risk score?

Family history of premature cardiovascular disease represents another important non-modifiable risk factor incorporated into the ASCVD-PREVENT score. This is typically defined as cardiovascular events occurring before age 55 in male first-degree relatives or before age 65 in female first-degree relatives. The presence of this factor can substantially increase a patient's calculated risk, reflecting the genetic component of cardiovascular disease susceptibility that may not be captured by other measurable risk factors.

Laboratory and Physical Findings

Laboratory parameters provide objective measures of physiological risk factors and are central to the ASCVD-PREVENT risk calculation. The lipid profile components—total cholesterol, LDL cholesterol, HDL cholesterol, and triglycerides—are key inputs that directly reflect atherogenic potential. Higher LDL cholesterol and triglyceride levels increase calculated risk, while higher HDL cholesterol levels have a protective effect in the algorithm, reducing the overall risk estimate.

Blood pressure measurements represent another critical physical finding incorporated into the risk score. The algorithm typically uses systolic blood pressure as the primary input, with higher values associated with increased cardiovascular risk. The score distinguishes between treated and untreated hypertension, recognizing that patients with treated hypertension may carry residual risk even when blood pressure is controlled. This distinction helps provide more accurat dy on antihypertensive therapy.

Additional laboratory findings that influence the ASCVD-PREVENT score calculation include:

Table

Parameter	Impact on Risk Score	Clinical Significance
High-sensitivity C-reactive protein (hsCRP)	Increases risk with higher values	Reflects systemic inflammation
Hemoglobin A1c	Increases risk with higher values	Indicates glycemic control in diabetics
Estimated glomerular filtration rate (eGFR)	Increases risk with lower values	Reflects kidney function
Lipoprotein(a)	Increases risk with higher values	Independent genetic risk factor

How Does the ASCVD–PREVENT Score Estimate Cardiovascular Risk in Dyslipidemia Patients? (LO#3)

The ASCVD-PREVENT score employs a sophisticated statistical model to translate patient-specific clinical inputs into a meaningful estimate of future cardiovascular risk. This model was developed through analysis of large-scale epidemiological data and clinical trials, incorporating both primary prevention populations (those without established cardiovascular disease) and secondary prevention populations (those with prior cardiovascular events). The resulting algorithm provides a percentage risk of experiencing a major adverse cardiovascular event within the next 10 years.

Risk Stratification

The ASCVD-PREVENT score stratifies patients into distinct risk categories based on their calculated 10-year risk percentage. These categories help clinicians contextualize the numerical risk estimate and guide appropriate management decisions.

The risk stratification categories for the ASCVD-PREVENT score are: (Figure 1)

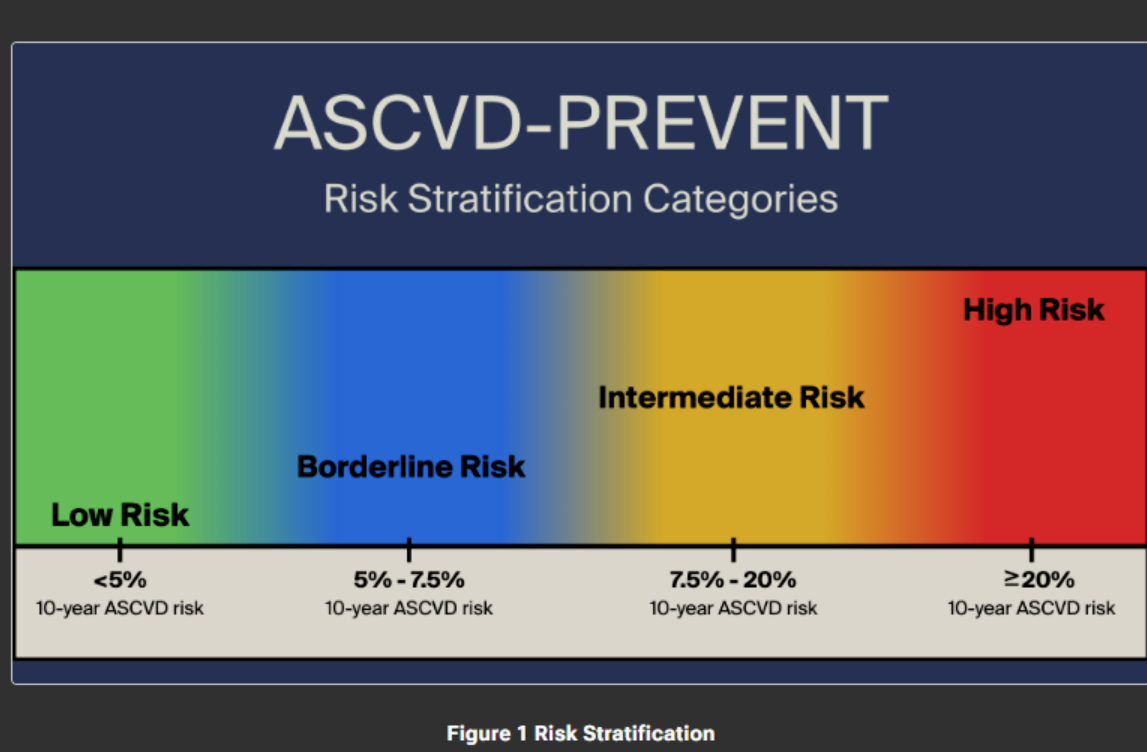

Figure 1 Risk Stratification

For patients with dyslipidemia, this stratification is particularly valuable as it helps determine the intensity of lipid-lowering therapy required. Patients in higher risk categories generally warrant more aggressive treatment approaches, includin potentially additional non-statin medications. The risk category also influences the target LDL cholesterol level, with more stringent targets typically recommended for higher-risk patients.

Interpretation of Scores

Interpreting ASCVD-PREVENT scores requires understanding both the absolute risk percentage and its implications for clinical management. The absolute risk percentage represents the estimated probability that a patient will experience a major adverse cardiovascular event (such as myocardial infarction, stroke, or cardiovascular death) within the next 10 years. This percentage should be considered alongside the patient's age, as baseline risk naturally increases with advancing age even in the absence of other risk factors.

Beyond the absolute risk percentage, the ASCVD-PREVENT score provides an estimate of the potential absolute risk reduction that could be achieved with statin therapy. This is calculated based on the patient's baseline risk and the expected relative risk reduction from appropriate statin therapy (typically around 25-35% for moderate-intensity statins and 35-45% for high-intensity statins). This feature helps clinicians and patients understand the potential benefit of therapy in absolute terms, facilitating more informed treatment decisions.

When interpreting scores for patients with dyslipidemia, it's important to consider several factors:

- Baseline LDL cholesterol level and its deviation from recommended targets
- Presence of additional risk factors that may modify the treatment approach
- Patient preferen n therapy
- Potential for risk reclassification with additional testing (such as coronary calcium scoring)

How Do You Use the ASCVD–PREVENT Risk Score to Calculate an Individual Patient's Cardiovascular Event Risk? (LO#3)

Calculating an individual patient's cardiovascular event risk using the ASCVD-PREVENT score requires a systematic approach to data collection and input into the risk calculator. While the underlying algorithm is complex, clinicians can utilize online calculators, electronic health record-integrated tools, or mobile applications to facilitate the calculation process. Understanding the step-by-step approach ensures accurate risk assessment and appropriate clinical application of the results.

Calculation Steps

The process of calculating the ASCVD-PREVENT risk score involves several sequential steps:

1. Gather all required patient information, including demographics, medical history, and current medications
2. Obtain necessary laboratory values, particularly a complete lipid panel (total cholesterol, LDL-C, HDL-C, triglycerides)
3. Measure and document the patient's blood pressure, preferably as an average of multiple readings
4. Input all collected data into an ASCVD-PREVENT risk calculator (available online or through electronic health record systems)
5. Review the calcu esponding risk category
6. Assess the estimated absolute risk reduction with statin therapy at different intensity levels

It's important to note that certain clinical scenarios may require additional consideration when calculating the ASCVD-PREVENT risk score. For patients already on statin therapy, the calculator can estimate what their untreated lipid levels might have been, though this introduces some uncertainty. For patients with very high LDL cholesterol levels (>190 mg/dL) or those with diabetes and age >40 years, high-intensity statin therapy is typically recommended regardless of the calculated risk score, as these conditions are considered high-risk equivalents.

Example Calculation

To illustrate the ASCVD-PREVENT risk calculation process, consider the following patient scenario:

Patient profile: 58-year-old non-Hispanic white male with no history of cardiovascular disease, non-smoker, with treated hypertension (current BP 138/82 mmHg), no diabetes, and the following laboratory values: total cholesterol 220 mg/dL, LDL-C 145 mg/dL, HDL-C 42 mg/dL, triglycerides 165 mg/dL.

Entering these values into the ASCVD-PREVENT calculator would generate the following results:

Table

Parameter	
10-year ASCVD risk	12.8%
Risk category	Intermediate risk
Estimated risk reduction with moderate-intensity statin	3.5% (absolute reduction)
Estimated risk reduction with high-intensity statin	5.1% (absolute reduction)
Recommended management	Consider moderate to high-intensity statin therapy

This example demonstrates how the ASCVD-PREVENT score translates multiple risk factors into an actionable risk estimate. For this patient, the 12.8% 10-year risk places him in the intermediate risk category, suggesting that statin therapy would likely be beneficial. The calculator also provides estimates of the absolute risk reduction with different statin intensities, allowing for a more informed discussion about the potential benefits of treatment.

What are the key steps in calculating the ASCVD-PREVENT risk score?

What are the clinical implications of using the ASCVD-PREVENT risk scores?

- *Lipid Management:* Helps determine the choice of pharmacological and non-pharmacological strategies for lowering lipid levels. For instance, for a high risk (≥20%) patient, recommend high-intensity statin therapy (unless contraindicated), aiming for ≥50% LDL-C reduction.
- *Blood Pressure Management:* Helps determine BP targets and whether pharmacotherapy beyond lifestyle measures is needed. High ASCVD risk strengthens the rationale for tighter BP control (<130/80 mmHg).
- *Diabetes Management:* In diabetic patients, ASCVD risk helps prioritize use of pharmacological agents, such as GLP-1 receptor agonists or SGLT2 inhibitors with proven cardiovascular benefit.
- *Aspirin for primary management:* Only consider in select adults at high ASCVD risk and low bleeding risk; the PREVENT score helps clarify net benefit.
- *Lifestyle Counseling:* Scores provide a quantitative way to motivate patients by showing the impact of lifestyle changes on long-term CVD outcomes.

The ASCVD-PREVENT risk score serves as a critical decision-making tool that guides clinicians in developing personalized treatment plans for patients with dyslipidemia. By quantifying cardiovascular risk and potential treatment benefits, the score helps determine appropriate therapeutic interventions, intensity of lipid-lowering therapy, and follow-up strategies. This risk-based approach ensures that treatment decisions align with the patient's individual risk profile rather than asurements.

Patient Counseling

Effective patient counseling is essential for translating ASCVD-PREVENT risk score results into meaningful behavior change and treatment adherence. When discussing risk score results with patients who have dyslipidemia, clinicians should focus on several key elements:

- Explaining the concept of 10-year risk in understandable terms
- Contextualizing the patient's risk relative to others of similar age and sex
- Illustrating the potential risk reduction with various interventions
- Addressing common concerns about statin therapy and other medications
- Emphasizing the importance of lifestyle modifications alongside pharmacotherapy

CLINICAL CORRELATION

Visual aids can be particularly helpful in communicating risk information effectively. Many ASCVD-PREVENT calculators provide graphical representations of risk and benefit, such as icon arrays showing the number of cardiovascular events that might be prevented with treatment. These visual tools help patients better understand abstract risk percentages and make more informed decisions about their care.

Follow-up planning is another important aspect of patient counseling that is guided by the ASCVD-PREVENT risk score. Higher-risk patients typically require more frequen ment of treatment response, while lower-risk patients may need less intensive follow-up. The score can help establish appropriate intervals for reassessment and guide discussions about when treatment intensification might be necessary if treatment goals are not achieved.

CASE CONNECTION

BACK TO INTRODUCTION ↑

Thinking back to RJ, what factors were necessary for calculating his ASCVD-PREVENT risk score? How would his individual risk estimate guide choices for managing his dyslipidemia?

For RJ, the 52-year-old male with elevated cholesterol, calculating his ASCVD-PREVENT risk score would require his age, sex, race/ethnicity, smoking status, blood pressure, diabetes status, total cholesterol, HDL cholesterol, LDL cholesterol, and current medications. His risk category would determine the appropriate intensity of statin therapy, with intermediate or high risk warranting moderate to high-intensity statins. The absolute risk reduction estimate would help RJ understand the potential benefit of treatment, facilitating shared decision-making about his dyslipidemia management.

Summary

- The ASCVD-PREVENT risk score estimates a patient's 10-year risk of atherosclerotic cardiovascular disease events and potential treatment benefits, particularly for dyslipidemia patients.
- The score incorporates non-modifiable factors (age, sex, race/ethnicity, family history), m physical activity, BMI), comorbidities (diabetes, hypertension, CKD), laboratory parameters (lipid panel, hsCRP), and current medications.
- ASCVD-PREVENT stratifies patients into risk categories (low, borderline, intermediate, high) to guide the intensity of lipid-lowering therapy and target LDL cholesterol levels.
- Calculating the ASCVD-PREVENT score involves collecting patient demographics, medical history, laboratory values, and blood pressure, then inputting data into a risk calculator to review the 10-year risk percentage and potential risk reduction with statin therapy.
- Management decisions for dyslipidemia are informed by ASCVD-PREVENT risk categories, with higher-risk patients typically requiring more intensive statin therapy and possibly additional non-statin medications.
- Effective patient counseling using ASCVD-PREVENT results involves explaining risk concepts, contextualizing individual risk, illustrating treatment benefits, addressing medication concerns, and emphasizing lifestyle modifications.

Review Questions

1. Which of the following patient variables is not directly included in the ASCVD-PREVENT risk score calculation?

A. ○ Patient's Age
B. ○ LDL cholesterol levels
C. ○ Diabetes status
D. ● Coronary
E. ○ Family history of CVD (Cardiovascular Disease)

Hide Explanation

The correct answer is Coronary calcium score (D). While coronary calcium scoring provides valuable information about subclinical atherosclerosis, it is not a standard input variable in the ASCVD-PREVENT risk score algorithm. Age (A) is a fundamental demographic factor in the calculation. LDL cholesterol (B) is a key laboratory parameter that directly influences risk estimation. Diabetes status (C) is an important comorbidity that significantly impacts cardiovascular risk assessment. Family history of cardiovascular disease (CVD) (E) is included as it reflects genetic predisposition to cardiovascular disease.

2. Which factor increases the ASCVD risk score the most?

A. ● Age
B. ○ Sex
C. ○ LDL levels
D. ○ Smoking
E. ○ Family history

Hide Explanation

The correct answer is age (A). Age is a powerful predictor of cardiovascular risk; individuals over 40 have higher weightings. While sex (B) affects risk, males have generally higher baseline risk but less than age. LDL levels (C) contribute but are less influential than age. Smoking (D) impacts risk but is not as determining as age. Family history (E) increases risk but less than age.

3. A 55-year-old male with dyslipidemia has a calculated ASCVD-PREVENT score of 15%. How would this patient's cardiovascular risk be categorized?

A. ○ Low risk
B. ○ Borderline risk
C. ● Intermediate risk
D. ○ High risk
E. ○ Very high risk

Hide Explanation

The correct answer is Intermediate risk (C). A 10-year ASCVD risk of 15% falls within the intermediate risk category, which is defined as 7.5% to <20%. Low risk (A) is defined as <5%, which is significantly lower than this patient's score. Borderline risk (B) encompasses the 5% to <7.5% range, below this patient's calculated risk. High risk (D) is defined as risk level. Very high risk (E) is not a standard category in the ASCVD-PREVENT risk stratification system.

(B) encompasses the 5% to <7.5% range, below this patient's calculated risk. High risk (D) is defined as ≥20%, which exceeds this patient's current risk level. Very high risk (E) is not a standard category in the ASCVD-PREVENT risk stratification system.

How would you rate this Brick?

Review Learning Objectives

Check off the objectives when you feel comfortable with the concept

01 ☑ ~~Define the purpose and components of the ASCVD-PREVENT risk score.~~
02 ☑ ~~Describe the clinical factors used in calculating the ASCVD-PREVENT risk score.~~
03 ☑ ~~Explain how the ASCVD-PREVENT risk score estimates cardiovascular risk in patients with dyslipidemia~~
04 ☑ ~~Apply the ASCVD-PREVENT risk score to calculate individual patient risk for cardiovascular events.~~

Objective Confidence: 4/4